利用功能進行備份複寫SnapVault

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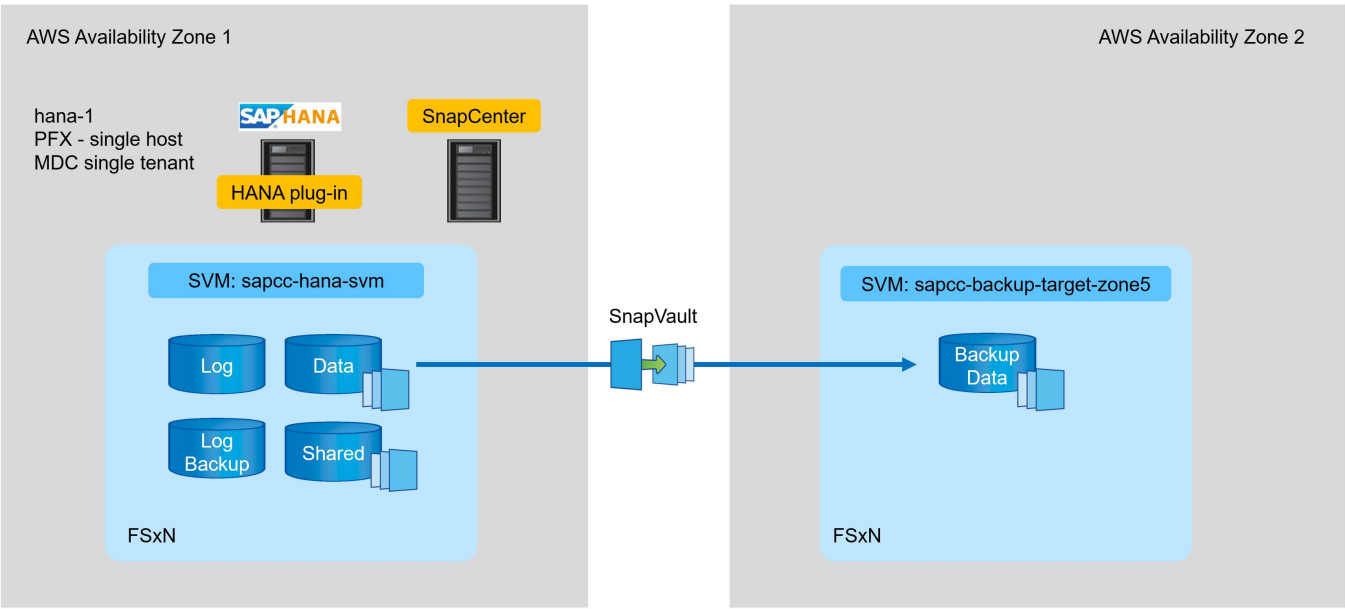
利用功能進行備份複寫SnapVault

總覽- SnapVault 利用功能進行備份複寫

"先前：非資料磁碟區的備份。"

在實驗室設定中、我們在ONTAP 第二個AWS可用度區域中使用第二個FSXfor the Sfor the Sfuse檔案系統來展示HANA資料Volume的備份複寫。

如第章所述 "「資料保護策略」"、複寫目標必須是ONTAP 另一個可用區域中的第二個FSXfor the Sf2檔案系統、才能避免ONTAP 發生適用於該文件系統的主要FSX故障。此外、HANA共享磁碟區也應複寫至次要FSXfor ONTAP the Sfor the Sf二 檔案系統。



組態步驟總覽

有幾個組態步驟必須在FSXfor ONTAP Sof the Sof the Layer上執行。您可以使用NetApp Cloud Manager或FSXfor ONTAP the Sfor the Command Line來執行此作業。

1. 對等FSXfor ONTAP Sf供 檔案系統使用。FSXfor ONTAP Sfor Sfor Sfor Sfor FeSffile系統必須進行對等處理、才能在彼此之間進行複寫。
2. 對等SVM：SVM必須進行對等處理、才能在彼此之間進行複寫。
3. 建立目標Volume。在目標SVM上建立磁碟區、磁碟區類型為「DP」。必須輸入「DP」作為複寫目標Volume。
4. 建立SnapMirror原則。這是用來建立複寫類型為「保存庫」的原則。
    a. 新增規則至原則。此規則包含SnapMirror標籤和保留資料、以便在次要站台進行備份。您必須稍後在SnapCenter 「還原」原則中設定相同的SnapMirror標籤、SnapCenter 以便在含有此標籤的來源Volume上建立Snapshot備份。
5. 建立SnapMirror關係。定義來源與目標Volume之間的複寫關係、並附加原則。
6. 初始化SnapMirror。這會啟動初始複寫、以便將完整的來源資料傳輸至目標Volume。

當Volume複寫組態完成時、您必須在SnapCenter 下列步驟中設定使用下列功能的備份複寫：

1. 將目標SVM新增SnapCenter 至SVM。
2. 為SnapCenter Snapshot備份和SnapVault 功能不全的複寫建立全新的功能不全原則。
3. 將原則新增至HANA資源保護。
4. 您現在可以使用新原則執行備份。

以下各章將更詳細地說明各個步驟。

"下一步：在FSX上設定ONTAP 適用於Sfor Sf供 檔案系統的複寫關係。"

在FSX上設定ONTAP 適用於Sfxfon檔案系統的複寫關係

"上一篇：概述- SnapVault 利用功能進行備份複寫。"

如需SnapMirror組態選項的詳細資訊、請參閱ONTAP 上的《關於SnapMirror組態選項的資訊》 "SnapMirror複寫工作流程（netapp.com）"。

- 資料來源FSX: ONTAP 「FsxId00fa9e3c784b6abbb」
- 來源SVM：「APCA-HA-SVM」
- 目標FSx用於ONTAP 支援資料系統：「FsxId05f7f00af49dc7a3e」
- 目標SVM：「APCPCC備份-目標-區域5」

對等FSXfor ONTAP Sf供 檔案系統使用

```
FsxId00fa9e3c784b6abbb::> network interface show -role intercluster
            Logical    Status     Network            Current       Current
Is
Vserver     Interface  Admin/Oper Address/Mask       Node          Port
Home
----------- ---------- ---------- ------------------ ------------- -------
----
FsxId00fa9e3c784b6abbb
            inter_1      up/up    10.1.1.57/24
FsxId00fa9e3c784b6abbb-01
                                                                   e0e
true
            inter_2      up/up    10.1.2.7/24
FsxId00fa9e3c784b6abbb-02
                                                                   e0e
true
2 entries were displayed.
```

```
FsxId05f7f00af49dc7a3e::> network interface show -role intercluster
            Logical    Status     Network            Current       Current
Is
Vserver     Interface  Admin/Oper Address/Mask       Node          Port
Home
----------- ---------- ---------- ------------------ ------------- -------
----
FsxId05f7f00af49dc7a3e
            inter_1      up/up    10.1.2.144/24
FsxId05f7f00af49dc7a3e-01
                                                                   e0e
true
            inter_2      up/up    10.1.2.69/24
FsxId05f7f00af49dc7a3e-02
                                                                   e0e
true
2 entries were displayed.
```

```
FsxId05f7f00af49dc7a3e::> cluster peer create -address-family ipv4 -peer
-addrs 10.1.1.57, 10.1.2.7
Notice: Use a generated passphrase or choose a passphrase of 8 or more
characters. To ensure the authenticity of the peering relationship, use a
phrase or sequence of characters that would be hard to guess.
Enter the passphrase:
Confirm the passphrase:
Notice: Now use the same passphrase in the "cluster peer create" command
in the other cluster.
```

「對等網路」是目的地叢集的叢集IP。

```
FsxId00fa9e3c784b6abbb::>  cluster peer create -address-family ipv4 -peer
-addrs  10.1.2.144, 10.1.2.69
Notice: Use a generated passphrase or choose a passphrase of 8 or more
characters. To ensure the authenticity of the peering relationship, use a
        phrase or sequence of characters that would be hard to guess.
Enter the passphrase:
Confirm the passphrase:
FsxId00fa9e3c784b6abbb::>
FsxId00fa9e3c784b6abbb::> cluster peer show
Peer Cluster Name         Cluster Serial Number Availability
Authentication
------------------------- --------------------- --------------
--------------
FsxId05f7f00af49dc7a3e    1-80-000011           Available      ok
```

對等SVM

```
FsxId05f7f00af49dc7a3e::> vserver peer create -vserver sapcc-backup-
target-zone5 -peer-vserver sapcc-hana-svm -peer-cluster
FsxId00fa9e3c784b6abbb -applications snapmirror
Info: [Job 41] 'vserver peer create' job queued
```

```
FsxId00fa9e3c784b6abbb::> vserver peer accept -vserver sapcc-hana-svm
-peer-vserver sapcc-backup-target-zone5
Info: [Job 960] 'vserver peer accept' job queued
```

```
FsxId05f7f00af49dc7a3e::> vserver peer show
            Peer        Peer                            Peering
Remote
Vserver     Vserver     State        Peer Cluster      Applications
Vserver
----------- ----------- ------------ ----------------- --------------
---------
sapcc-backup-target-zone5
            peer-source-cluster
                        peered       FsxId00fa9e3c784b6abbb
                                                       snapmirror
sapcc-hana-svm
```

建立目標Volume

您必須使用「DP」類型建立目標磁碟區、將其標示為複寫目標。

```
FsxId05f7f00af49dc7a3e::> volume create -vserver sapcc-backup-target-zone5
-volume PFX_data_mnt00001 -aggregate aggr1 -size 100GB -state online
-policy default -type DP -autosize-mode grow_shrink -snapshot-policy none
-foreground true -tiering-policy all -anti-ransomware-state disabled
[Job 42] Job succeeded: Successful
```

建立SnapMirror原則

SnapMirror原則和新增規則定義了保留和SnapMirror標籤、以識別應複寫的Snapshot。稍後建立SnapCenter 「樣」原則時、您必須使用相同的標籤。

```
FsxId05f7f00af49dc7a3e::> snapmirror policy create -policy snapcenter-
policy -tries 8 -transfer-priority normal -ignore-atime false -restart
always -type vault -vserver sapcc-backup-target-zone5
```

```
FsxId05f7f00af49dc7a3e::> snapmirror policy add-rule -vserver sapcc-
backup-target-zone5  -policy snapcenter-policy -snapmirror-label
snapcenter -keep 14
```

```
FsxId00fa9e3c784b6abbb::> snapmirror policy showVserver Policy
Policy Number         Transfer
Name    Name               Type   Of Rules Tries Priority Comment
------- ------------------ ------ -------- ----- -------- ----------
FsxId00fa9e3c784b6abbb
        snapcenter-policy  vault         1     8  normal  -
  SnapMirror Label: snapcenter                        Keep:      14
                                                Total Keep:      14
```

建立SnapMirror關係

現在已定義來源與目標Volume之間的關係、以及XDP類型和我們先前建立的原則。

```
FsxId05f7f00af49dc7a3e::> snapmirror create -source-path sapcc-hana-
svm:PFX_data_mnt00001 -destination-path sapcc-backup-target-
zone5:PFX_data_mnt00001 -vserver sapcc-backup-target-zone5 -throttle
unlimited -identity-preserve false -type XDP -policy snapcenter-policy
Operation succeeded: snapmirror create for the relationship with
destination "sapcc-backup-target-zone5:PFX_data_mnt00001".
```

初始化**SnapMirror**

使用此命令會啟動初始複寫。這是將所有資料從來源磁碟區完整傳輸至目標磁碟區的方式。

```
FsxId05f7f00af49dc7a3e::> snapmirror initialize -destination-path sapcc-
backup-target-zone5:PFX_data_mnt00001 -source-path sapcc-hana-
svm:PFX_data_mnt00001
Operation is queued: snapmirror initialize of destination "sapcc-backup-
target-zone5:PFX_data_mnt00001".
```

您可以使用「napmirror show」命令來檢查複寫狀態。

```
FsxId05f7f00af49dc7a3e::> snapmirror show

Progress
Source            Destination Mirror  Relationship   Total
Last
Path        Type  Path        State   Status         Progress  Healthy
Updated
----------- ---- ------------ ------- -------------- --------- -------
--------
sapcc-hana-svm:PFX_data_mnt00001
            XDP  sapcc-backup-target-zone5:PFX_data_mnt00001
                             Uninitialized
                                     Transferring   1009MB    true
02/24 12:34:28
```

```
FsxId05f7f00af49dc7a3e::> snapmirror show

Progress
Source            Destination Mirror  Relationship   Total
Last
Path        Type  Path        State   Status         Progress  Healthy
Updated
----------- ---- ------------ ------- -------------- --------- -------
--------
sapcc-hana-svm:PFX_data_mnt00001
            XDP  sapcc-backup-target-zone5:PFX_data_mnt00001
                             Snapmirrored
                                     Idle           -         true    -
```

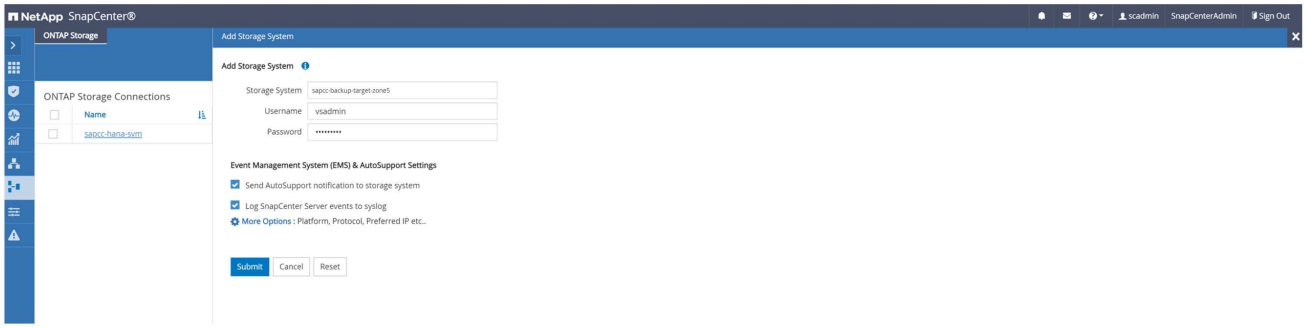
"下一步：將備份SVM新增SnapCenter 至"

將備份SVM新增SnapCenter 至

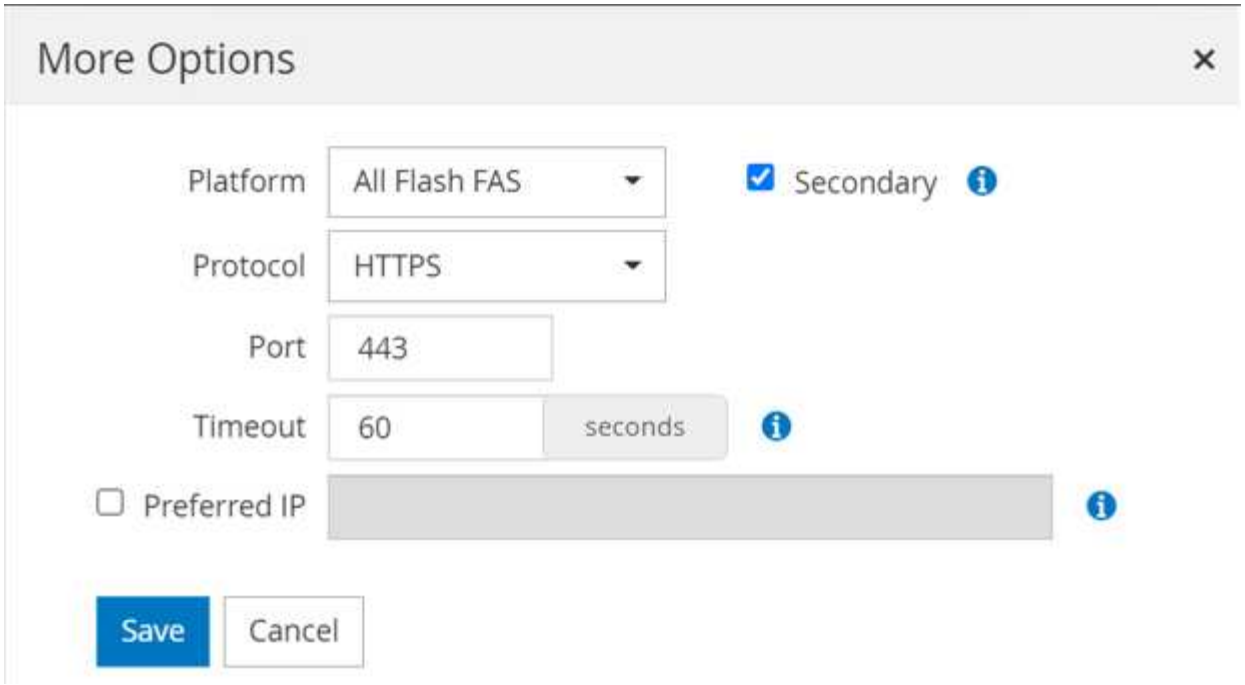
"上一篇：在FSX上 設定ONTAP 適用於Sfxfon檔案系統的複寫關係。"

若要將備份SVM新增SnapCenter 至SVM、請依照下列步驟操作：

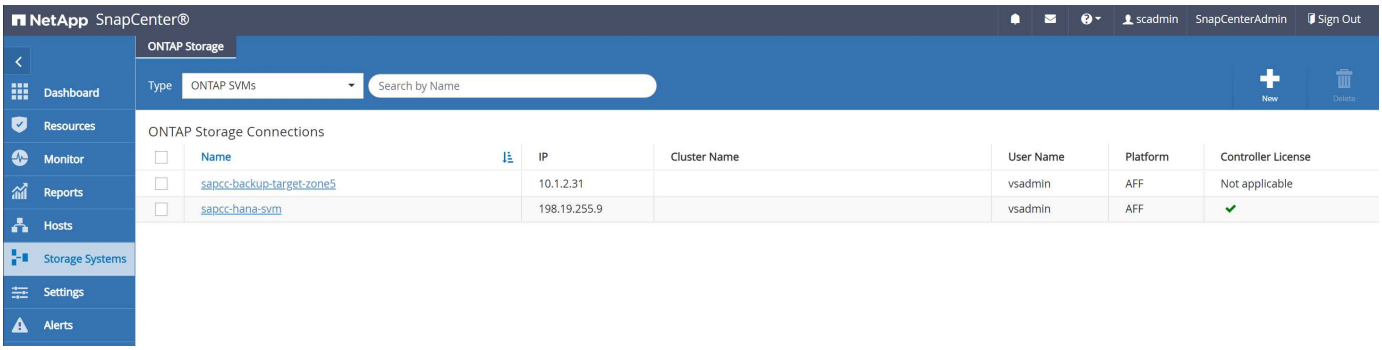
1. 設定SVM、SnapVault 使其位於SVM SnapCenter 中的SVM磁碟區。



2. 在「More Options（更多選項）」視窗中、選取「All Flash FAS 」作為平台、然後選取「Secondary（次要）」。



SVM現已推出SnapCenter 以供選購。



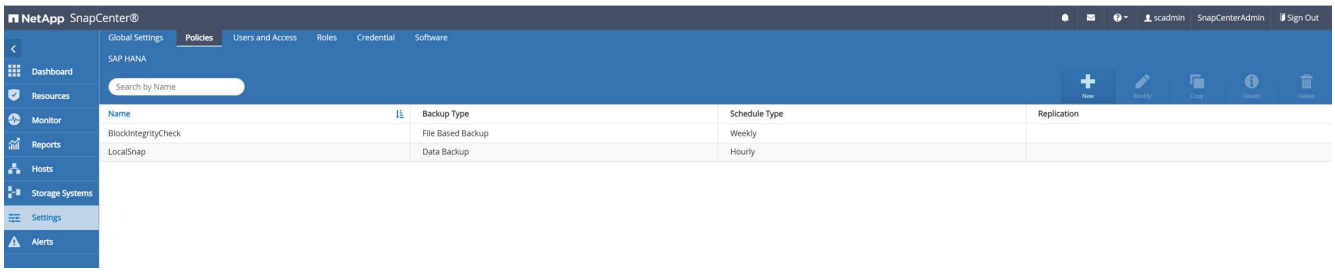
"下一步：建立SnapCenter 新的支援複寫功能的更新功能。"

建立SnapCenter 新的支援複寫功能的更新功能

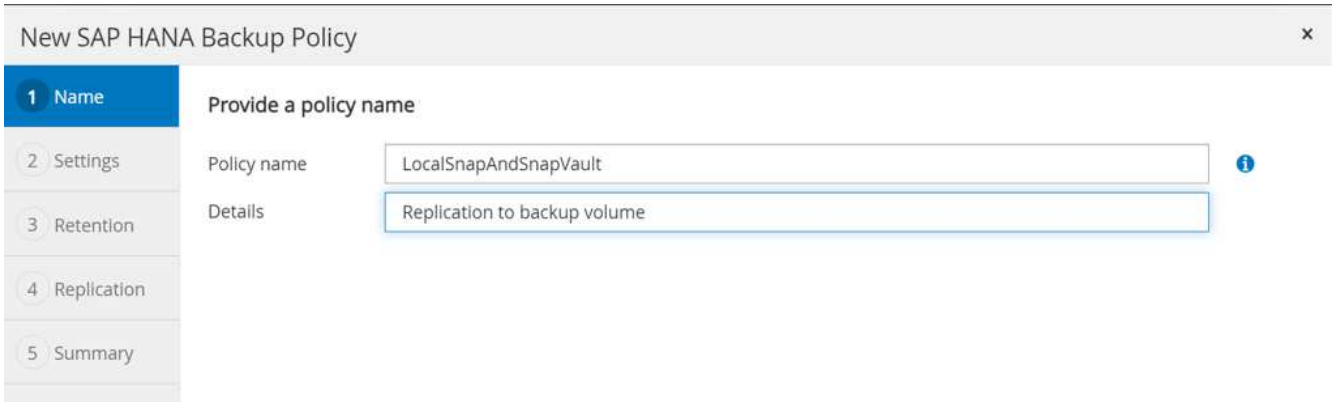
"上一步：將備份SVM新增SnapCenter 至SVM。"

您必須設定備份複寫原則、如下所示：

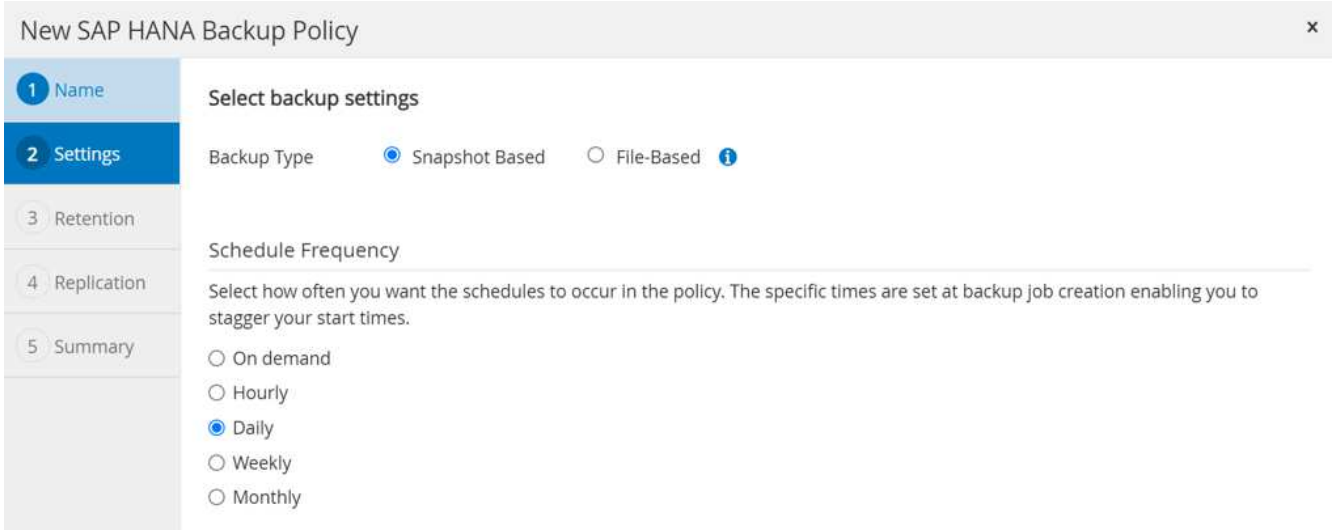
1. 提供原則名稱。



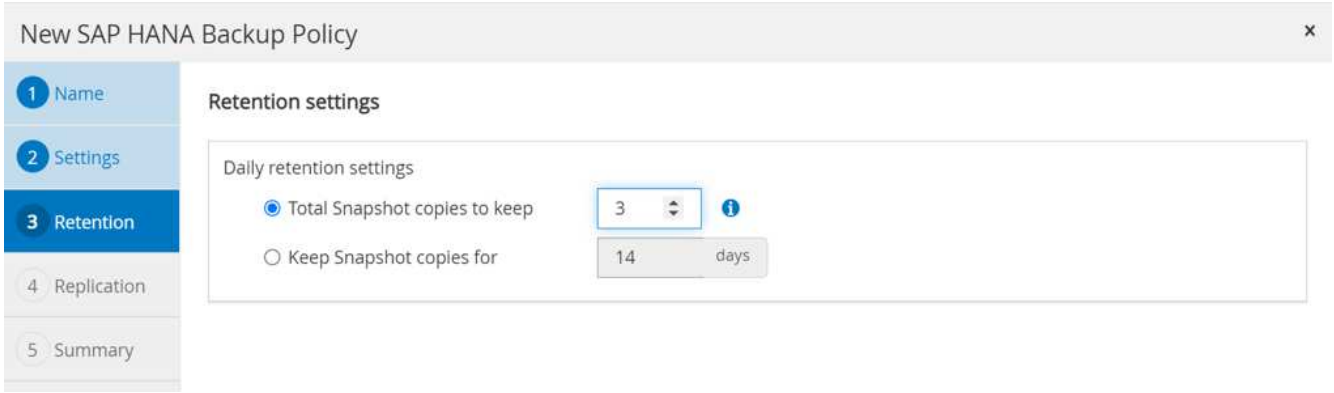
2. 選取Snapshot備份和排程頻率。每日通常用於備份複寫。



3. 選取Snapshot備份的保留。

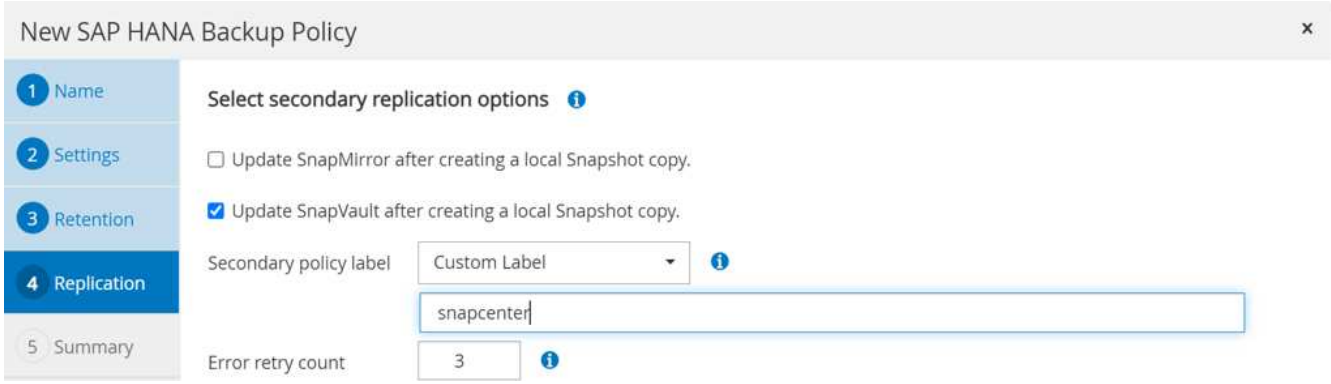


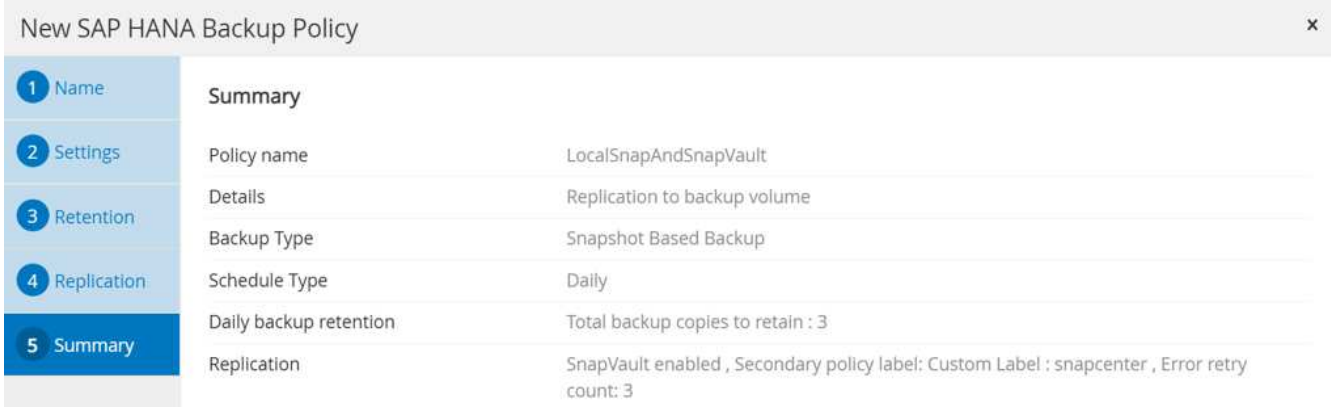
這是主儲存設備每日執行Snapshot備份的保留時間。先前已使用位於該層級的add rule命令、在還原目標中設定次要備份的保留。SnapVault ONTAP請參閱「在FSx上設定ONTAP 適用於不支援的檔案系統的複寫關係」（xref）。



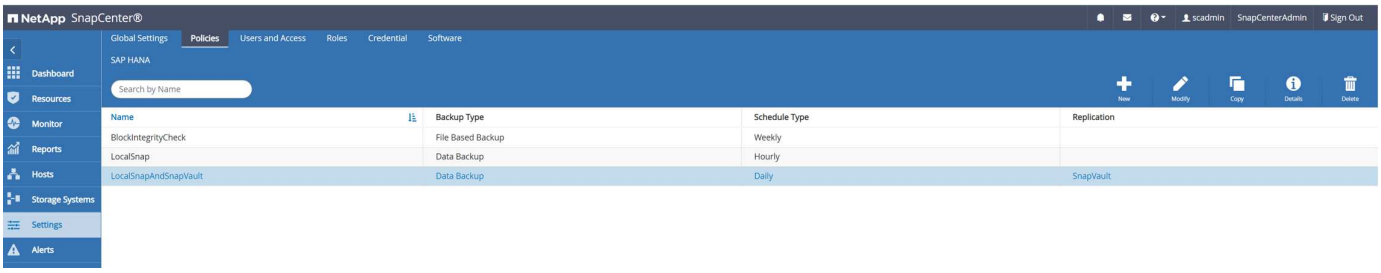
4. 選取「更新SnapVault 」功能變數、並提供自訂標籤。

此標籤必須符合ONTAP 「ADD規則」命令所提供的SnapMirror標籤、位於景點的層級。





現在SnapCenter 已設定新的功能更新原則。

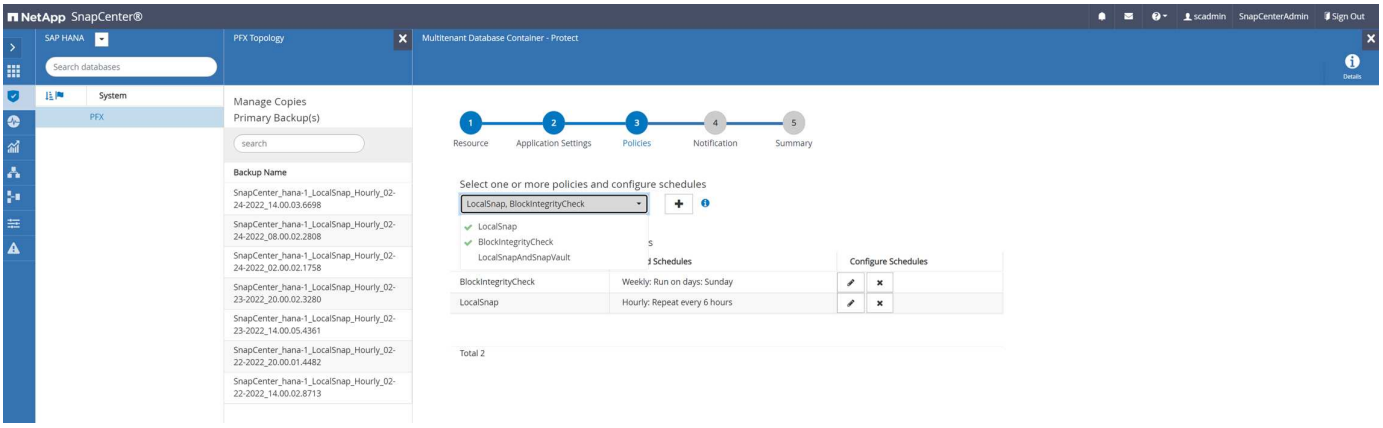


"下一步：新增原則至資源保護。"

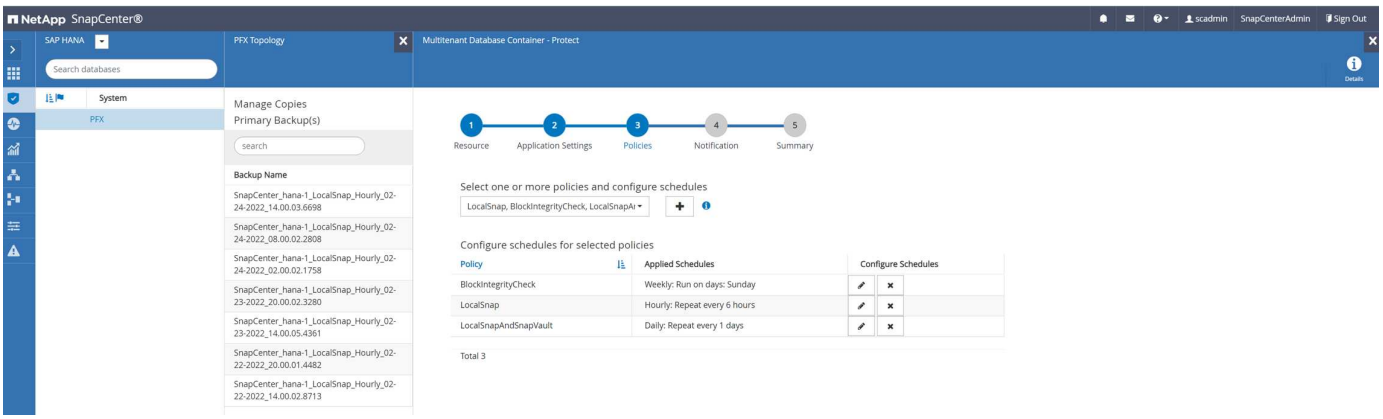
新增原則至資源保護

"上一篇：建立SnapCenter 新的支援複寫的更新政策。"

您必須將新原則新增至HANA資源保護組態、如下圖所示。



我們的設定定義了每日排程。



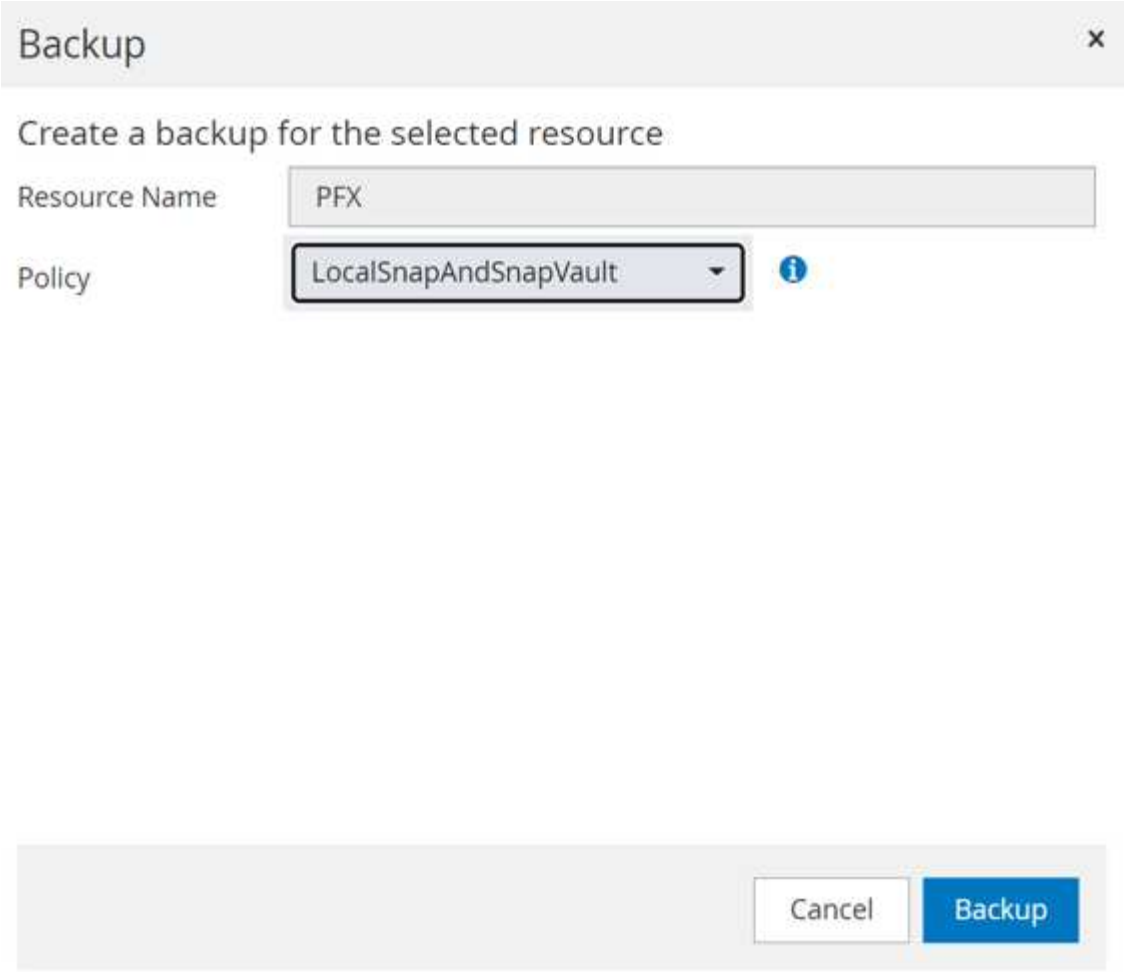
"下一步：建立複寫備份。"

使用複寫建立備份

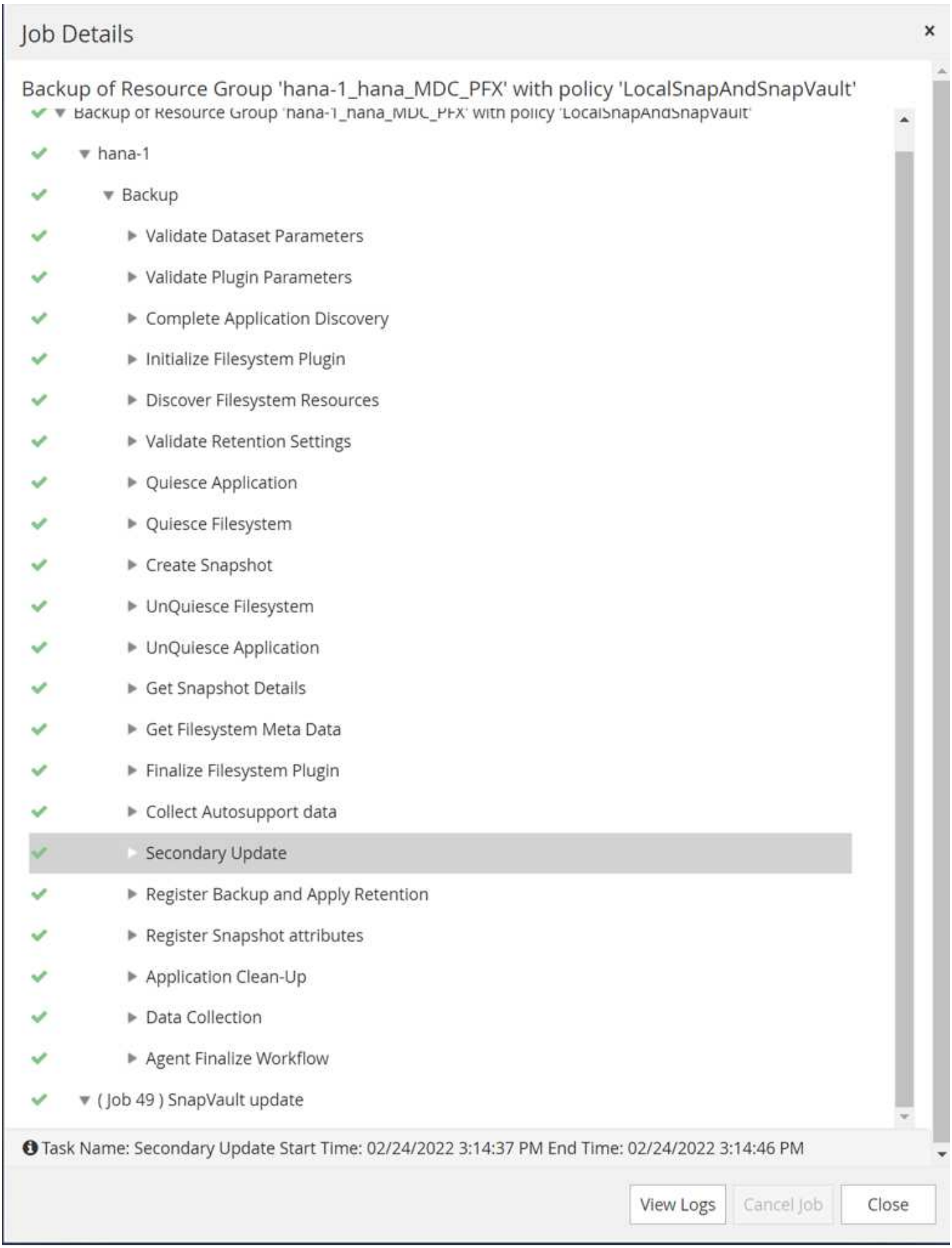
"上一步：新增原則至資源保護。"

備份的建立方式與本機Snapshot複本相同。

若要建立複寫備份、請選取包含備份複寫的原則、然後按一下備份。



在「職務記錄」中SnapCenter 、您可以看到「次要更新」步驟、這會啟動SnapVault 「更新」作業。複寫作業會將來源磁碟區的區塊變更為目標磁碟區。



在FSXfor ONTAP Sfor Sf2檔案系統上、會使用SnapCenter SnapMirror標籤「Sfor the Sf2」建立來源磁碟區上

的Snapshot、此標籤是SnapCenter 根據《支援資料》原則所設定的「支援資料」。

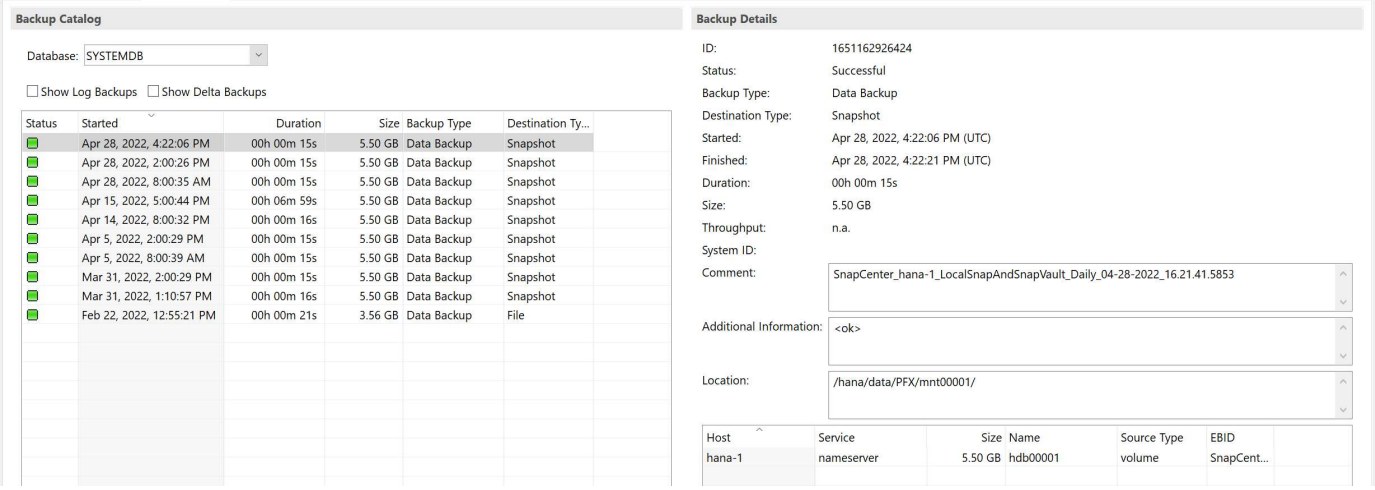
```
FsxId00fa9e3c784b6abbb::> snapshot show -vserver sapcc-hana-svm -volume
PFX_data_mnt00001 -fields snapmirror-label
vserver        volume            snapshot
snapmirror-label
-------------- -----------------
------------------------------------------------------------
----------------
sapcc-hana-svm PFX_data_mnt00001 SnapCenter_hana-1_LocalSnap_Hourly_03-31-
2022_13.10.26.5482 -
sapcc-hana-svm PFX_data_mnt00001 SnapCenter_hana-1_LocalSnap_Hourly_03-31-
2022_14.00.05.2023 -
sapcc-hana-svm PFX_data_mnt00001 SnapCenter_hana-1_LocalSnap_Hourly_04-05-
2022_08.00.06.3380 -
sapcc-hana-svm PFX_data_mnt00001 SnapCenter_hana-1_LocalSnap_Hourly_04-05-
2022_14.00.01.6482 -
sapcc-hana-svm PFX_data_mnt00001 SnapCenter_hana-1_LocalSnap_Hourly_04-14-
2022_20.00.05.0316 -
sapcc-hana-svm PFX_data_mnt00001 SnapCenter_hana-1_LocalSnap_Hourly_04-28-
2022_08.00.06.3629 -
sapcc-hana-svm PFX_data_mnt00001 SnapCenter_hana-1_LocalSnap_Hourly_04-28-
2022_14.00.01.7275 -
sapcc-hana-svm PFX_data_mnt00001 SnapCenter_hana-
1_LocalSnapAndSnapVault_Daily_04-28-2022_16.21.41.5853

snapcenter
8 entries were displayed.
```

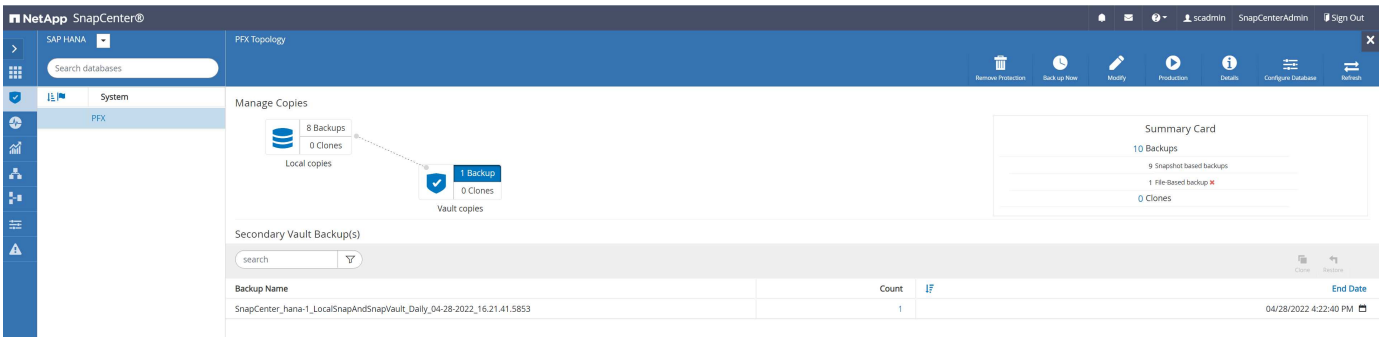
在目標磁碟區上、會建立名稱相同的Snapshot複本。

```
FsxId05f7f00af49dc7a3e::> snapshot show -vserver sapcc-backup-target-zone5
-volume PFX_data_mnt00001 -fields snapmirror-label
vserver                   volume            snapshot
snapmirror-label
------------------------- -----------------
-------------------------------------------------------------------------
----------------
sapcc-backup-target-zone5 PFX_data_mnt00001 SnapCenter_hana-
1_LocalSnapAndSnapVault_Daily_04-28-2022_16.21.41.5853 snapcenter
FsxId05f7f00af49dc7a3e::>
```

新的Snapshot備份也會列在HANA備份目錄中。



在還原中SnapCenter 、您可以按一下拓撲檢視中的Vault複本、以列出複寫的備份。



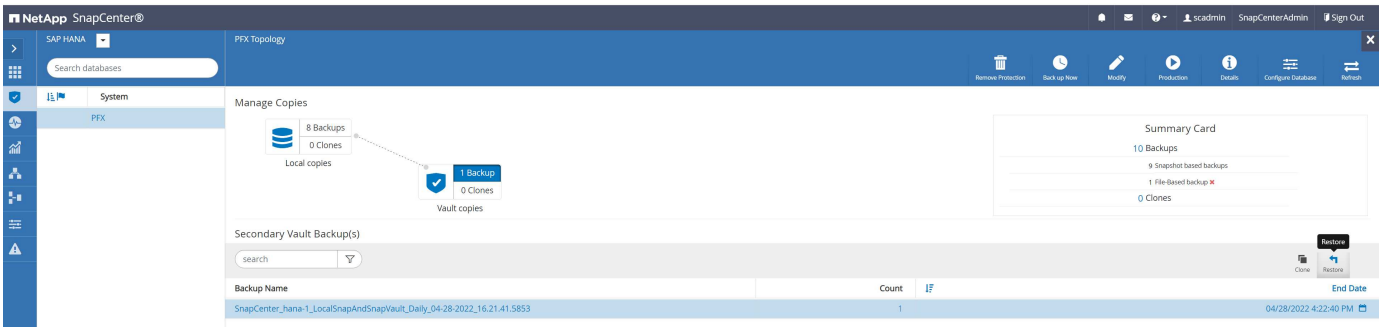
"下一步：從二線儲存設備還原及還原。"

從二線儲存設備還原及還原

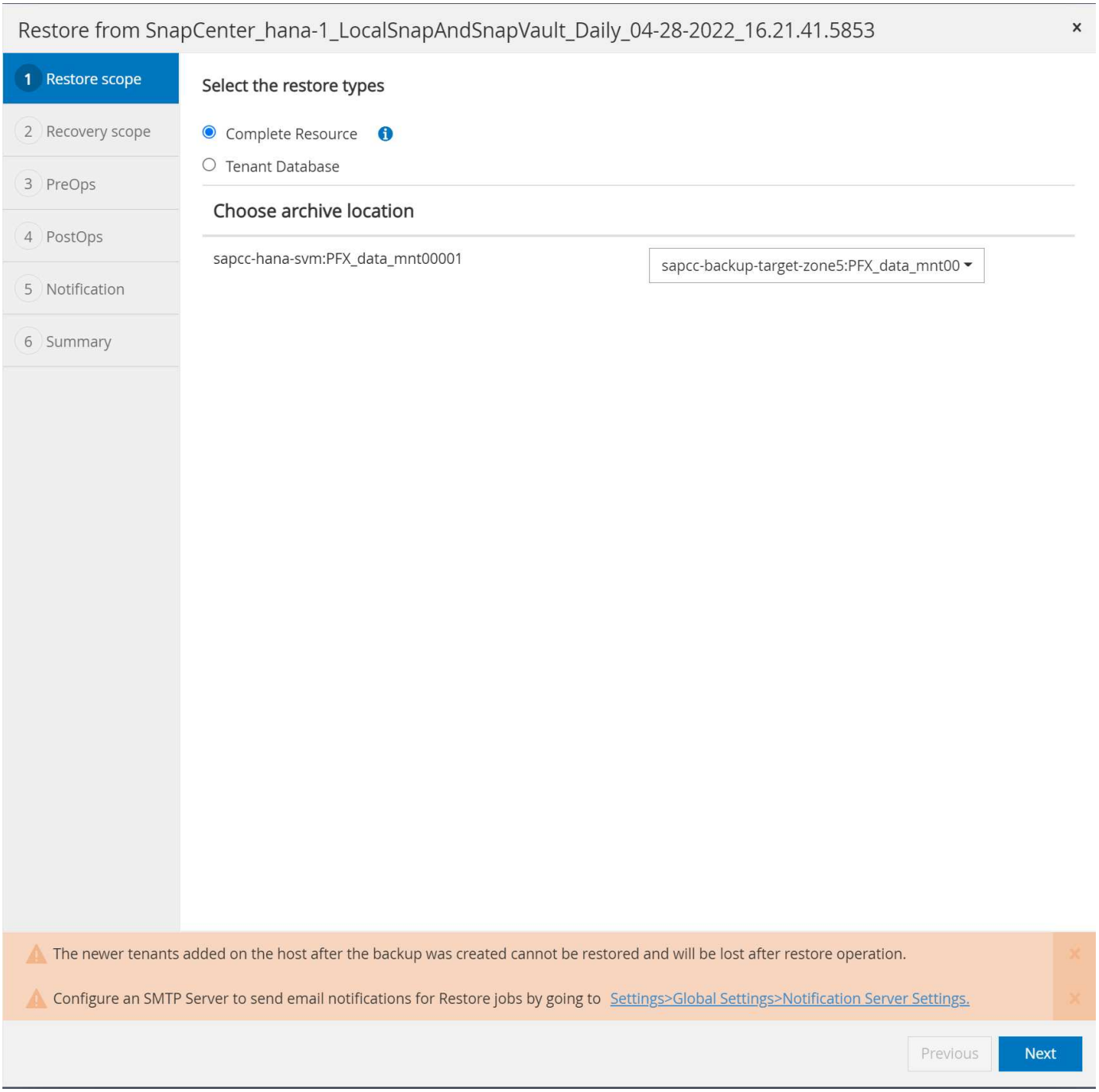
"上一篇：使用複寫建立備份。"

若要從次要儲存設備還原及還原、請依照下列步驟進行：

若要擷取次要儲存設備上所有備份的清單、請SnapCenter 在「還原拓撲」檢視中、按一下「Vault複本」、然後選取備份、再按一下「還原」。



還原對話方塊會顯示次要位置。



進一步的還原和還原步驟與先前在主要儲存設備上進行Snapshot備份所涵蓋的步驟相同。

"下一步：何處可以找到其他資訊和版本歷程記錄。"

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